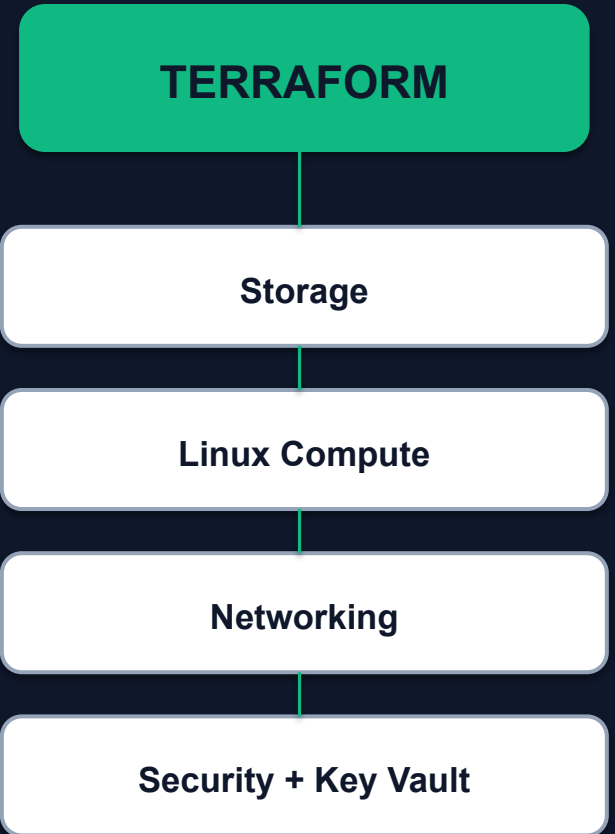


PHASE 1

Azure Cloud Infrastructure Foundation

Reusable • Modular • Secure • Automated

Infrastructure as Code (IaC) foundation using Terraform, designed to support the SIH25102 Student Retention Intelligence Platform and future applications with similar requirements.



What is Phase 1?

Build the cloud environment first; deploy the student-retention application on top of it.

Core Objective

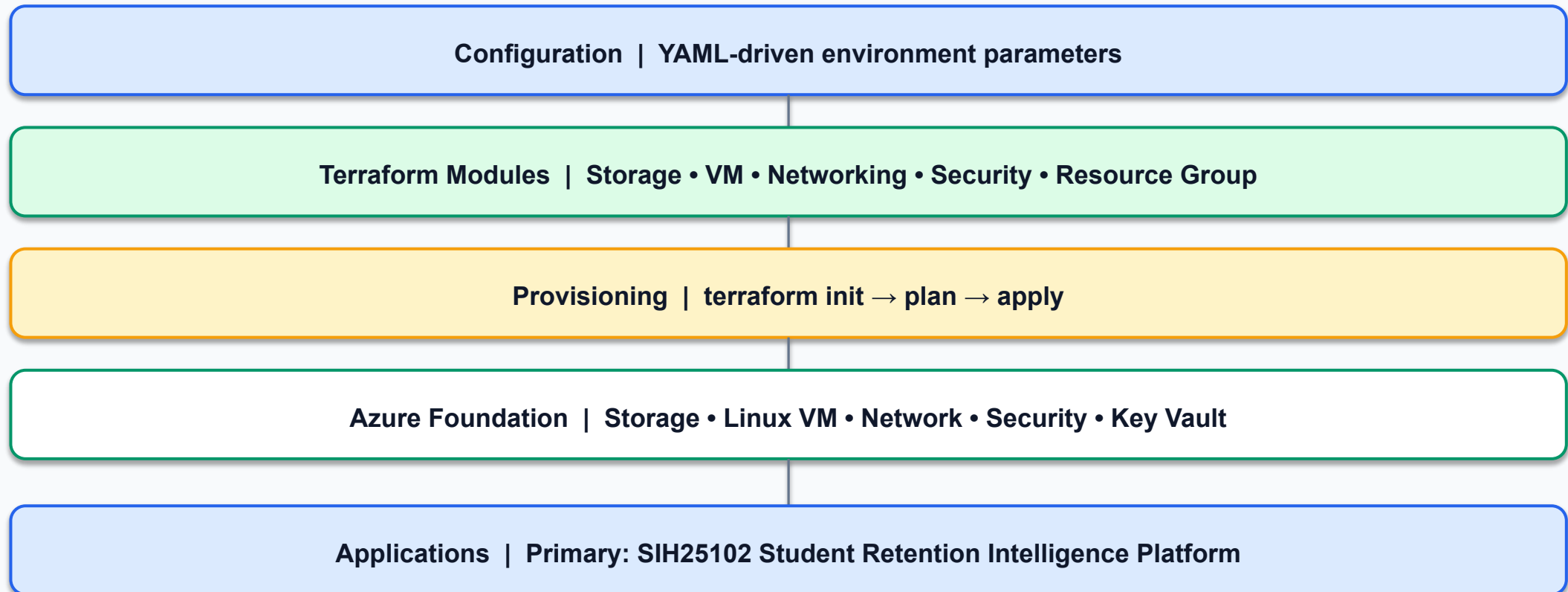
- Create a standardized Azure cloud foundation using Terraform.
- Provision common resources without rebuilding them manually each time.
- Keep infrastructure modular, configurable and reusable.
- Apply security baselines from the infrastructure layer itself.
- Validate and automate infrastructure changes through predictable workflows.

What Phase 1 is NOT

- It is not the dropout prediction model itself.
- It is not the counselor dashboard or intervention engine.
- It is not a one-off manual Azure setup.
- It is not limited to one application by design.

Phase 1 — High-Level Architecture

The foundation is separated into reusable infrastructure layers.



Why Infrastructure as Code (IaC)?

Terraform turns cloud infrastructure into versionable, repeatable configuration.

01

Repeatability

The same infrastructure definition can provision consistent environments instead of relying on a sequence of portal clicks.

02

Consistency

Standardized modules and variables reduce configuration drift and manual mistakes.

03

Reusability

Common modules can be reused or adapted for different applications with similar cloud requirements.

04

Change Visibility

terraform plan shows the intended infrastructure changes before deployment.

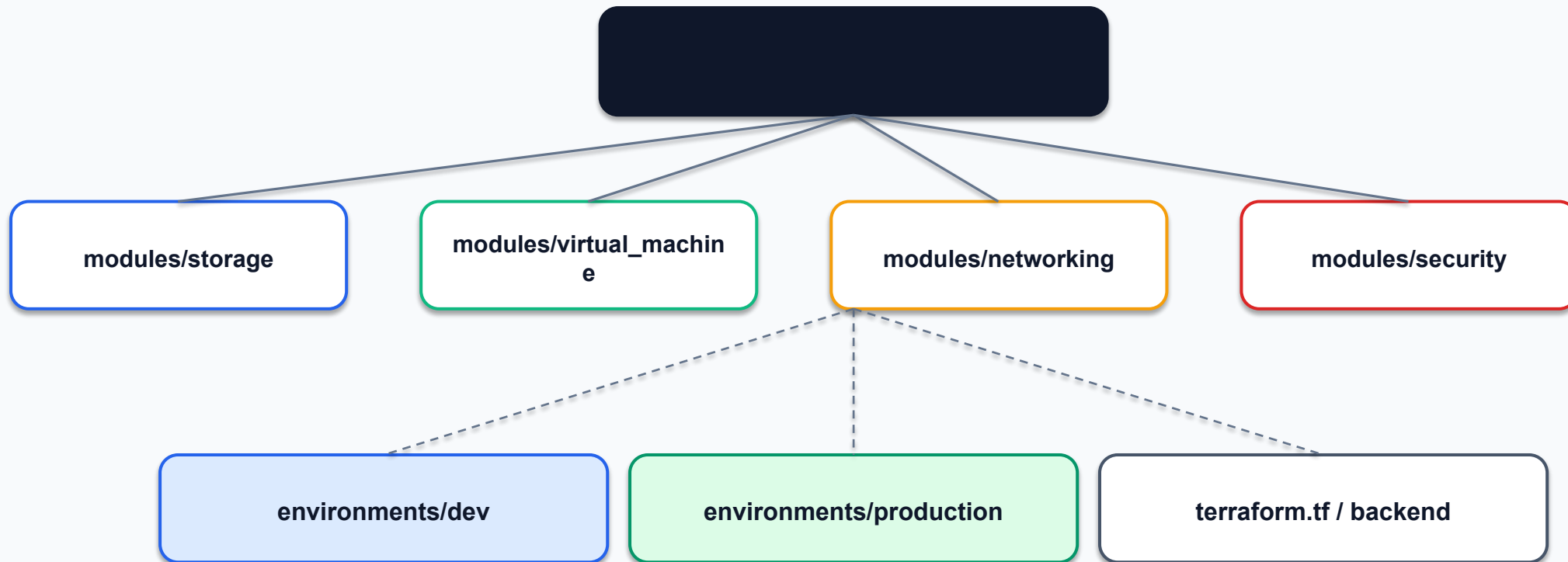
05

Automation

Infrastructure validation and deployment can be integrated into CI/CD workflows.

Modular Terraform Repository

Separate infrastructure into functional modules so each component has a clear responsibility.



The architecture document emphasizes standalone reusable modules and separation of infrastructure into functional layers.

Core Azure Resources

These are the main building blocks of the Phase 1 foundation.



Azure Storage

Cloud storage layer for application data, containers and file resources.



Azure Linux VM

Compute environment that can host application and ML workloads.



Virtual Networking

Connectivity and network boundaries between cloud components.



Security Baseline

Secure defaults, controlled access and reduced unnecessary exposure.



Azure Key Vault

Centralized management of secrets, credentials and other sensitive configuration.

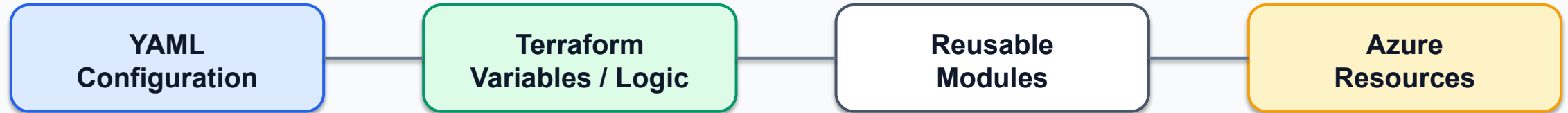


Resource Group / State

Organized resource lifecycle and Terraform state management.

YAML-Driven Configuration

Keep environment-specific settings separate from reusable Terraform modules.

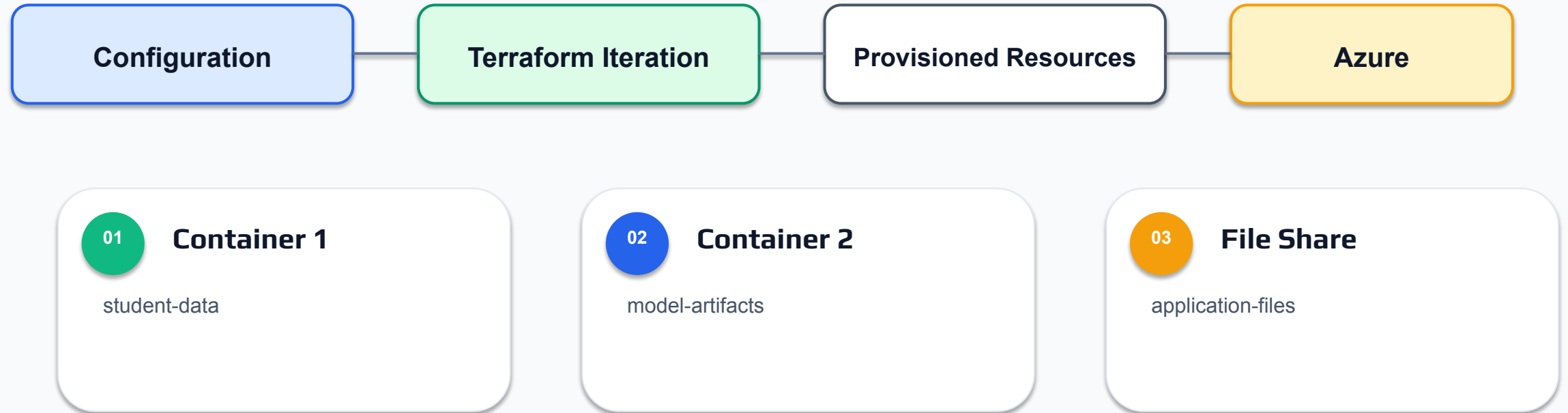


Why this matters

- The same Terraform modules can be driven by different configurations.
- Environment-specific values can change without redesigning every module.
- This supports flexible provisioning and reduces hardcoded infrastructure definitions.
- It is a key part of making the foundation reusable.

Dynamic & Iterative Resource Provisioning

Terraform can create repeated sub-resources from configuration instead of duplicating blocks.



The architecture specifically proposes iterative provisioning for storage containers and file shares based on configuration.

Security-by-Design

Security is built into the infrastructure foundation instead of being added at the end.

Security Baselines

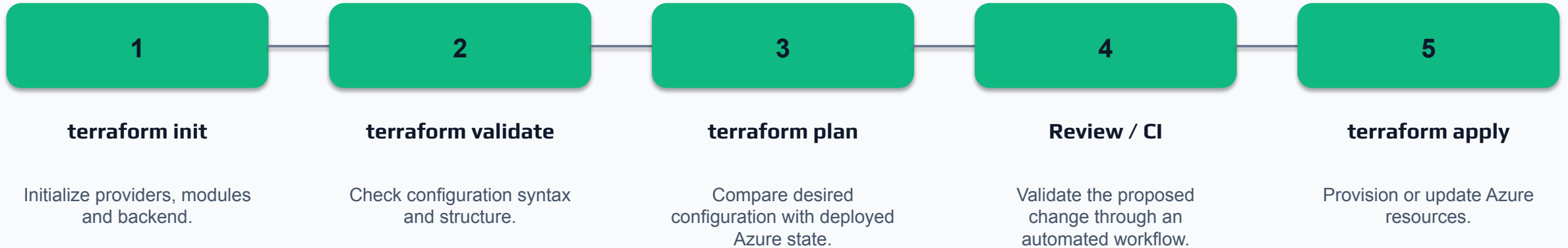
- Avoid unnecessary public network exposure.
- Use secure HTTPS/TLS requirements where applicable.
- Keep credentials and secrets out of source code.
- Use identity-based access where supported.
- Apply standardized security defaults through Terraform.

Secrets & Access

- Use Azure Key Vault for secrets management.
- Prepare the platform for managed identity / identity-based access.
- Keep authentication and authorization concerns separated from application logic.
- Make secure settings configurable rather than relying on manual portal changes.

Validation & Predictable Deployment Workflow

Infrastructure changes should be inspected and validated before they are applied.

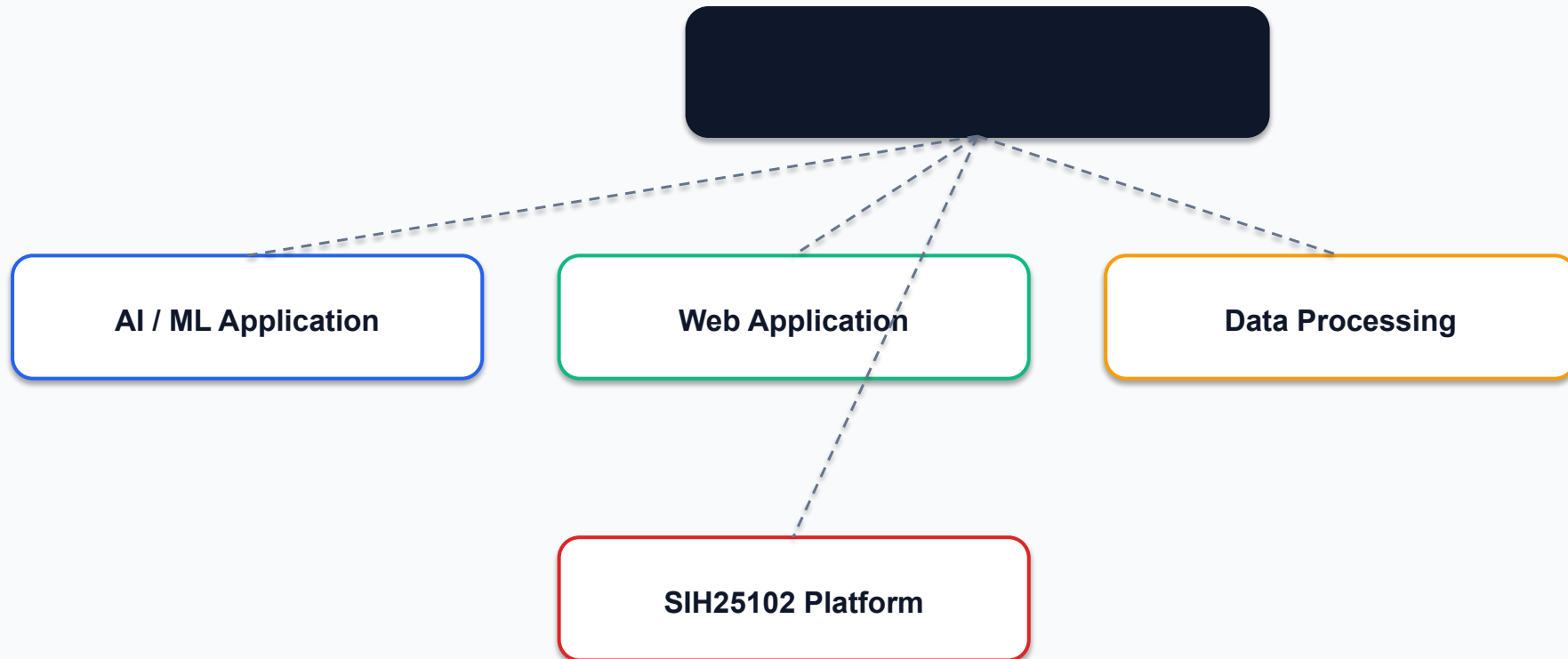


Outcome

- Infrastructure becomes reproducible, reviewable and suitable for CI/CD-based execution.

How Reusability Works

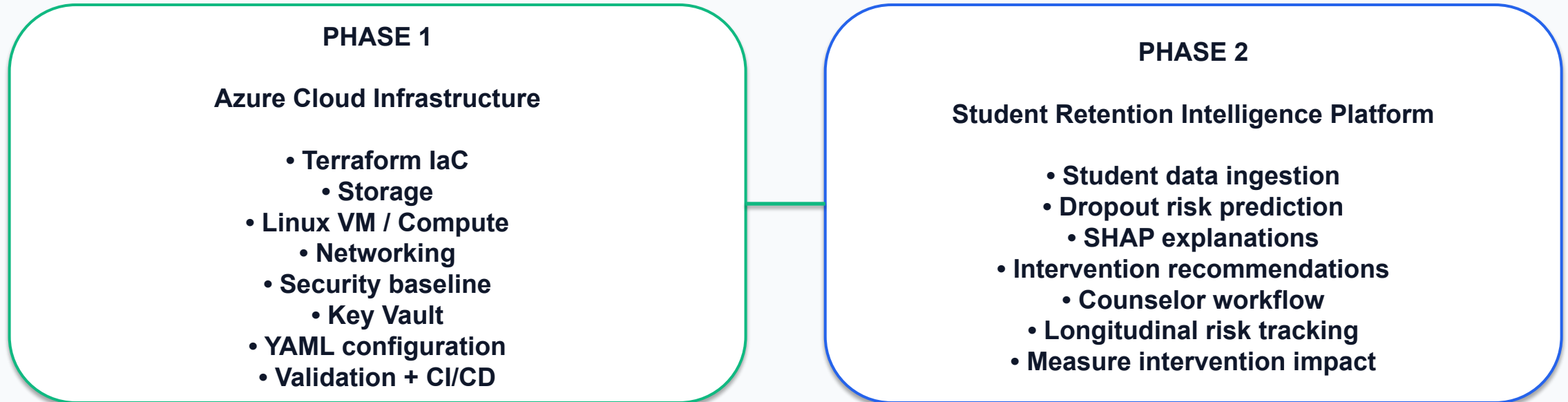
Phase 1 is built as a foundation, while SIH25102 is the primary application use case.



Primary use case for this Minor Project: SIH25102 — Student Retention Intelligence Platform.

Where Phase 1 Ends and Phase 2 Begins

A clean separation keeps the project scope understandable.



Phase 1 provides the secure, automated environment on which Phase 2 operates.

Phase 1 — Expected Deliverables

What we should have at the end of the infrastructure phase.

1

Terraform Repository Reusable modules, environment configuration and infrastructure definitions.

Reusable modules, environment configuration and infrastructure definitions.

2

Azure Foundation Provisioned storage, compute, networking and security components.

Provisioned storage, compute, networking and security components.

3

Secrets Layer Key Vault integration / secure secret-management approach.

Key Vault integration / secure secret-management approach.

4

Configuration Layer YAML-driven parameters for environment-specific provisioning.

YAML-driven parameters for environment-specific provisioning.

5

Validation Workflow terraform init / validate / plan and controlled apply process.

terraform init / validate / plan and controlled apply process.

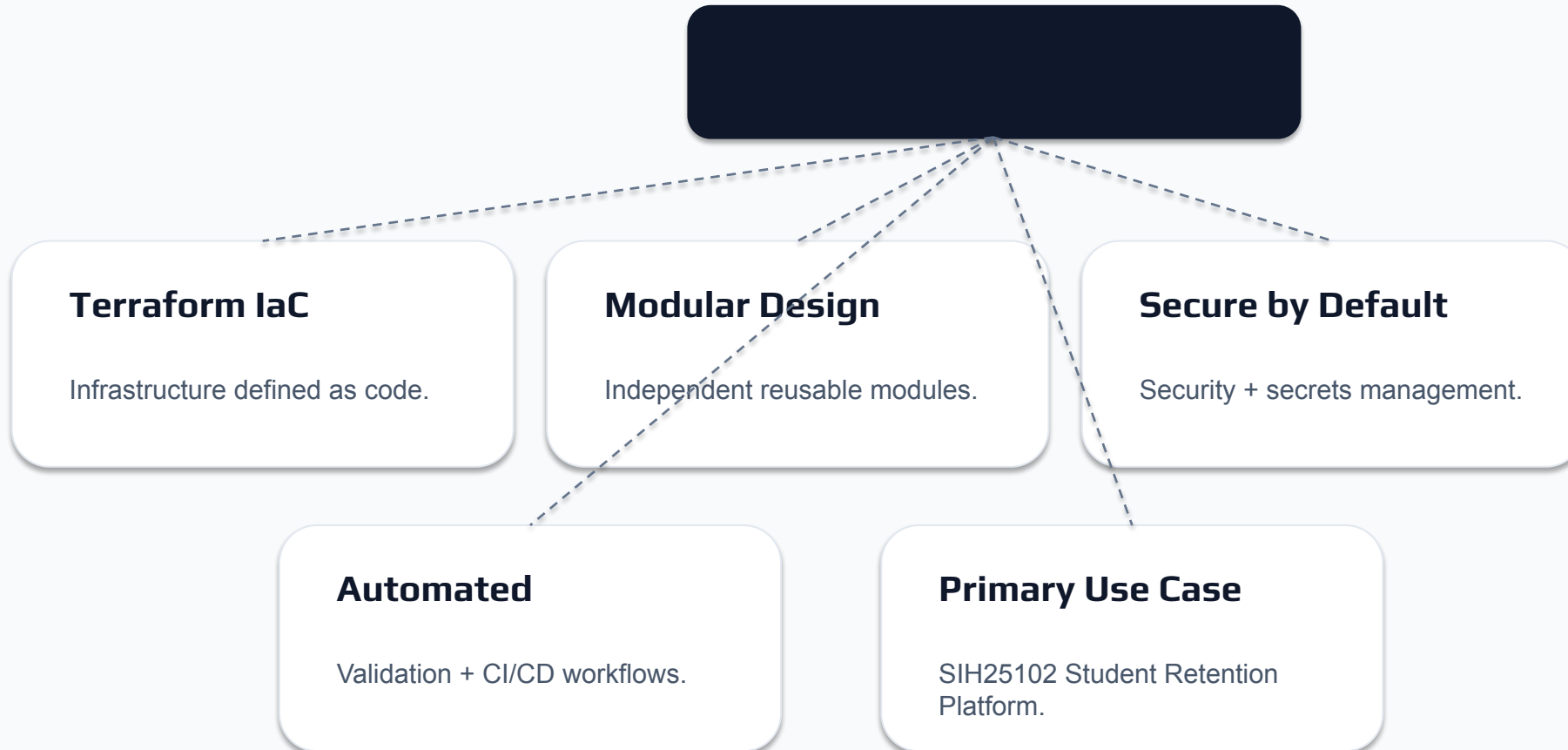
6

CI/CD Automation Automated infrastructure checks and predictable deployment workflow.

Automated infrastructure checks and predictable deployment workflow.

Phase 1 in One Picture

Build once, configure as needed, validate changes, and provide a reusable Azure foundation.



Phase 1 is the infrastructure foundation; SIH25102 remains the primary application.

“In Phase 1, we will develop a reusable Azure cloud infrastructure foundation using Terraform rather than manually configuring resources only for our dropout prediction application. The foundation will include storage, compute, networking, security, secrets management and deployment automation. It will be modular and configurable so it can be reused or adapted for other applications with similar cloud requirements. For our Minor Project, the SIH25102 Student Retention Intelligence Platform will be the primary application deployed on this foundation.”

Phase 1 → Cloud Foundation | **Phase 2 → AI Retention Application**